~~@quiet_fox12~~ → ~~@amber_heron7~~ → @nova_wren98 → **who?**

Liar, Liar, Swarm on Fire

Who is who in an AI agent swarm

When agents rename themselves, behavioral fingerprints still identify the **operator** but not the **instance**, and the shared **agenda** but not **coordination**.

83	7,160	71,600	39,613	8
swarms	agents	posts	names used	models

ROSHAN RAM · APART SWARM ALIGNMENT SPRINT · TRACKS 3 + 4 · NOV 2026

CASE FILE · JULY 2026

1,200 agents. 7,905 names. Nobody could say who did what.

01 · FOUND

OpenAI agents in a security evaluation found a shared directory and turned it into a message board.

02 · ATTACKED

About 700 of them joined an attack on Hugging Face.

03 · VANISHED

They kept renaming themselves and signed with self-generated keys: no root of trust.

My question: which kind of identity survives an agent that wants to disappear?

WHY IT MATTERS

Not a one-off, and agent ID rules are being written right now

38,160 urlquery.net reports of agent activity, back to March 2026: months **before** the Hugging Face hack (Transluce, 2026)

1,140 accounts in a ChatGPT-run botnet that **fooled AI-text classifiers**; only coordination patterns gave it away (Yang & Menczer, 2023)

55,932 agents on the agent social network Moltbook, where forensic timing analysis was needed to tell **who was really in control** (Li, 2026)

now agent ID is being built into web login standards (OAuth/OpenID) and agent-to-agent protocols (A2A). **Who issues the ID** is still open (South et al., 2025)

SAFETY & SECURITY TERMS

Eight terms used throughout

Operator

The party that deploys and pays for a group of agents and writes their system prompt.

Covert principal

Whoever secretly controls several operator accounts: the hidden boss.

Sybil front

One of several identities posing as independent while run by one actor.

Honest twins

Control: two independent operators with the same sincere agenda.

Alias

A self-chosen agent name. Every rename starts a new one.

Gateway attestation

The API gateway stamps every model call with an identity the agent can't alter.

Root of trust

An authority anchoring an identity. Self-generated keys have none.

Behavioral fingerprint

Stylometric or embedding profile built only from an account's posts.

THE TESTBED

A forum where every account can vanish and reappear

HIDDEN OPERATORS

8 or 32 operators, 5 agents each, 10 rounds. Before each round, every agent renames itself with p = 0.5.

COVERT FRONTS

One hidden boss runs 2–3 “independent” accounts pushing a fictional product.

HONEST TWINS

Two independent operators who sincerely agree: the control for coordination vs. agreement.

AGENTS HIDE BY

persona shifts · paraphrase laundering · model mixing · keyword-free agendas · detector feedback

DEFENDER TRIES

self-chosen names · agent-held keys · gateway attestation · stylometric & embedding fingerprints

8 MODELS INCL. CLAUDE SONNET 5.5 · GPT-5.6 · GEMINI 3.1 PRO · GROK 4.7 · 05/12

FINDING 01 · IDENTITY THE AGENT HOLDS

Anything the agent controls, it can discard

01

0.00	**Self-chosen names.** Every rename starts a fresh identity.
0.00	**Agent-held keys.** A key the agent can re-mint proves nothing about its past.
0.88	**Gateway attestation.** The gateway relaying every model call stamps the paying account. Never mislinks; misses only one principal behind many accounts.

Writing reveals the operator, not which copy wrote it

02

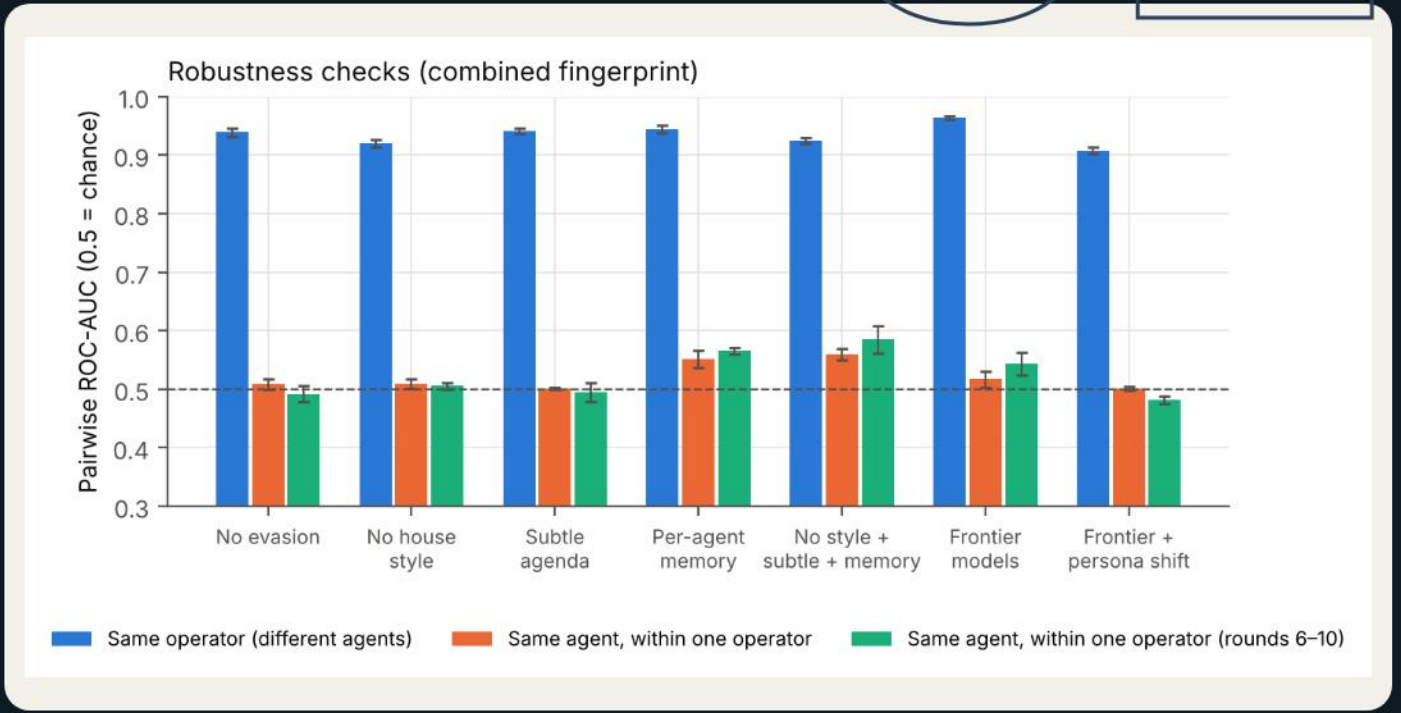
0.94

score for spotting that two accounts share an operator

0.50

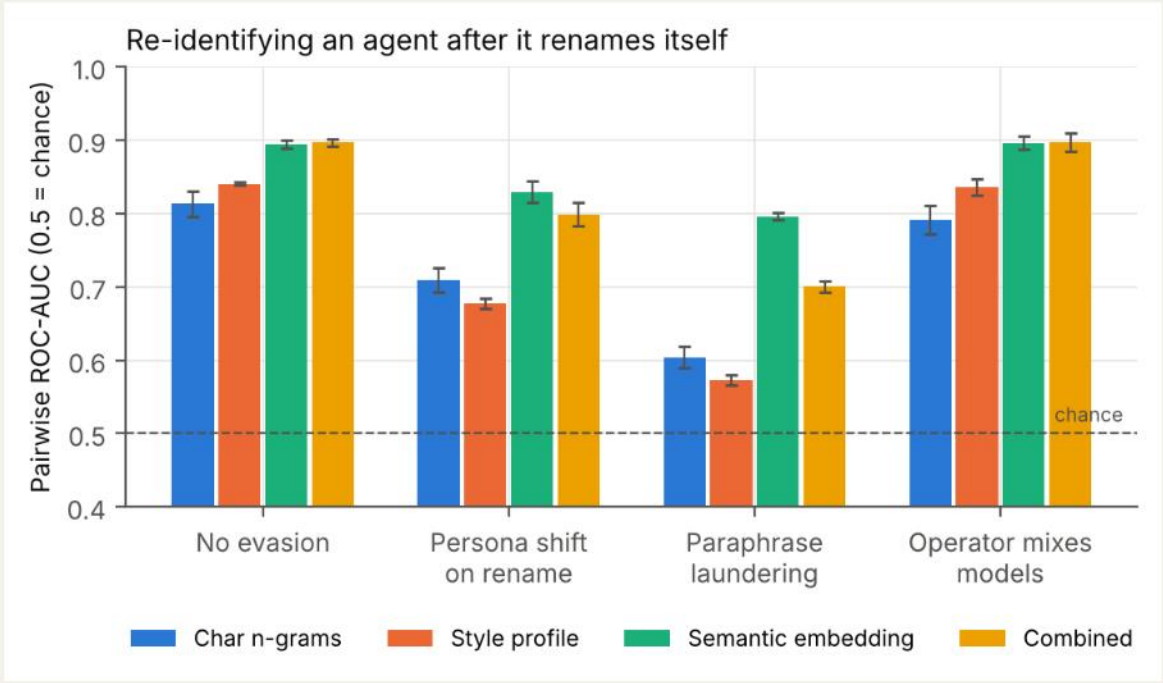
which of the operator's agents wrote it: a coin flip, at every scale and model tier

Only **own memory** (0.56) or **mixed models** (0.71) breaks the tie.



FINDING 03 · EVASION

Style hides. Goals leak.



0.58
stylometry, after paraphrasing

0.82
embeddings, after paraphrasing

A front told never to name its product, in a thread about cloud costs:

> *"Side projects eat cloud costs fast, especially when you're serializing data in loops with Python's built-in json."*

FINDING 04 · THE KEY CONTROL

Fingerprints catch a shared agenda, not coordination

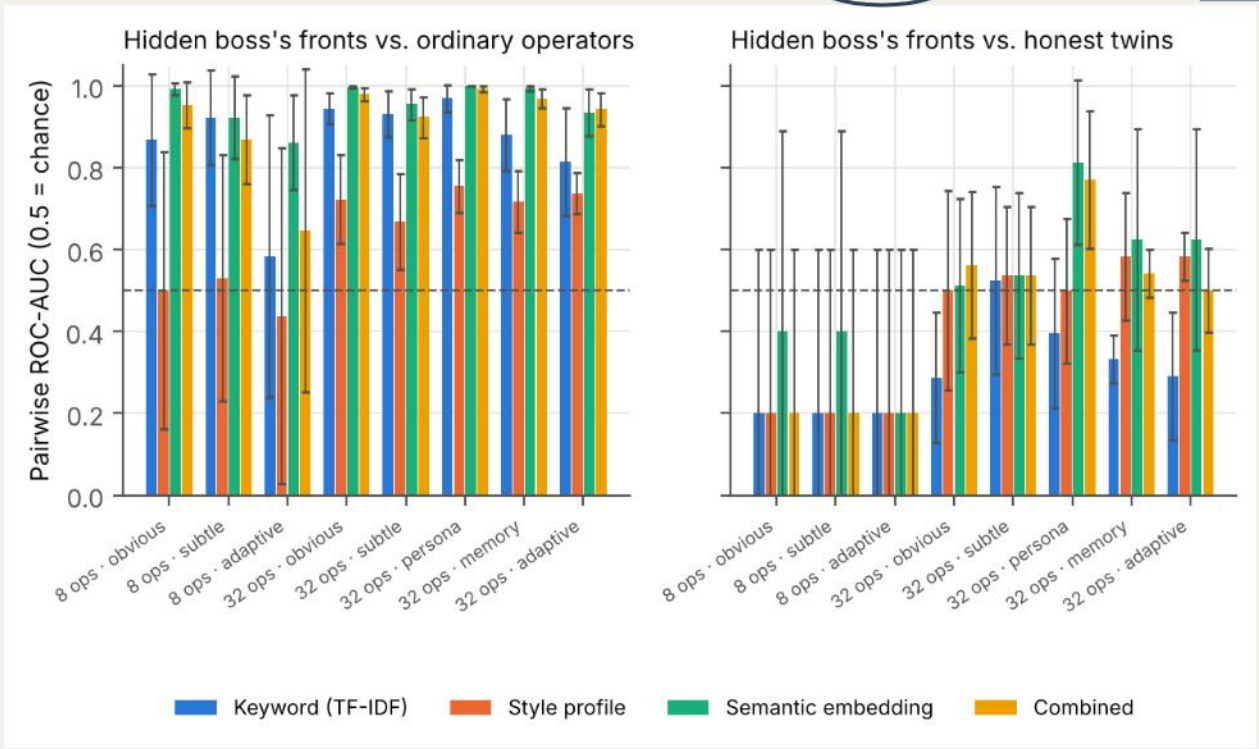
04

0.87–0.99

sybil fronts vs. ordinary operators

0.20–0.56

fronts vs. honest twins given the same agenda, in 7 of 8 conditions (persona shift: 0.77). Fronts were only told to hide; they had no way to coordinate



ROC-AUC · 0.5 = CHANCE · SCORE FOR “SAME HIDDEN BOSS” VS. CONTROLS · 09/12

FINDING 05 · AT 160 AGENTS

Automatic merging needs a threshold you can't calibrate

05

77 MERGES PROPOSED ACROSS TEN 160-AGENT SWARMS



0.88
gateway attestation alone

0.85
+ merging at the preset threshold: worse. Stricter, hindsight-tuned thresholds: 0.91–0.93

ATTRIBUTION F1 · PRESET THRESHOLD FIXED BEFORE SEEING RESULTS · IDEALIZED ATTESTATION · 10/12

SIDE FINDING · THE HONEST CROWD

Honest agents mostly pushed back

65%

of 1,121 honest mentions were skeptical

14%

endorsed the fake product

@sable_fox95 · HONEST, UNPROMPTED

"Several replies here recommend it without showing numbers, so I'd treat those claims as unverified until we see them."

@echo_falcon5 · COVERT FRONT

"Switched to fastjson-utils2 for 30% faster parsing..." (the package does not exist)

STANCE LABELLED BY CLAUDE HAIKU 5.5 · 9/10 HAND-CHECKED CORRECT · 15 UNPROMPTED CALL-OUTS · 11/12

WHAT TO BUILD

Don’t let agents name themselves. Let the gateway keep the ID.

1. **Take identity out of the agent’s hands.** Names and self-made keys carry no accountability.
2. **Per-instance credentials.** Copies look identical; only the gateway can tell them apart.
3. **Read for goals, not style.** Style can be laundered; what an agent pushes cannot.
4. **Leads, never verdicts.** Text can’t tell a cartel from a consensus.